Collaborative Network for Super-Resolution and Semantic Segmentation of Remote Sensing Images

Qian Zhang, Guang Yang, and Guixu Zhang

***Abstract*— In the past few years, multitask learning (MTL) has been widely used in a single model to solve the problems of multiple businesses. MTL enables each task to achieve high performance and greatly reduces computational resource overhead. In this work, we designed a collaborative network that simultaneously solves the super-resolution semantic segmentation and super-resolution image reconstruction. This algorithm can obtain high-resolution semantic segmentation and super-resolution reconstruction results by taking relatively low-resolution images as input when high-resolution data are inconvenient or computing resources are limited. The framework consists of three parts: the semantic segmentation branch (SSB), the super-resolution branch (SRB), and the structural affinity block (SAB). Specifically, the SSB, SRB, and SAB are responsible for completing super-resolution semantic segmentation, image super-resolution reconstruction, and associated features, respectively. Our proposed method is simple and efficient, and it can replace the different branches with most of the state-of-the-art models. The International Society for Photogrammetry and Remote Sensing (ISPRS) segmentation benchmarks were used to evaluate our models. In particular, super-resolution semantic segmentation on the Potsdam dataset reduced Intersection over Union (IoU) by only 1.8% when the resolution of the input image was reduced by a factor of two. The experimental results showed that our framework can obtain more accurate semantic segmentation and super-resolution reconstruction results than the single model.**

***Index Terms*— Multitask learning (MTL), remote sensing, semantic segmentation, super resolution.**

I. Introduction

SEMANTIC segmentation and super-resolution reconstruction of remote sensing images are basic tasks in computer vision. They refer to assigning a category tag to each pixel of remote sensing images and using relatively low-resolution remote sensing images to reconstruct high-resolution remote sensing images.

With the development of artificial intelligence (AI), the method represented by deep convolutional neural networks (DCNNs) has greatly improved the results of image semantic segmentation. However, the existing semantic segmentation models with high precision all depend on the input image with high resolution. On the one hand, the quality of existing low-resolution remote sensing images is too low to meet the learning requirements of DCNNs. On the other hand, high-resolution remote sensing images are not only expensive to acquire but also have a great demand for computing resources when applied in DCNNs. Acquiring high-quality semantic segmentation images and super-resolution image reconstruction by using relatively low-resolution images through DCNNs is challenging.

In the field of remote sensing, different from ordinary camera imaging, images taken by airborne or space-borne equipment are typically affected to different degrees by factors [1] such as ground sampling area [2], atmospheric attenuation [3], and atmospheric path radiation [4]. Obtaining high-quality high-resolution images is difficult. As a solution, a single image super-resolution can be used to reconstruct a low-resolution image into a high-resolution one. Combining single image super-resolution and semantic segmentation, the low-resolution image can be used as input to obtain a high-resolution segmentation image.

A. Semantic Segmentation

Image semantic segmentation is an important branch in the AI field. It refers to the use of DCNNs to learn pixel-to-pixel mapping, automatic segmentation, and recognition of the content in the image [5]. Before the development of deep learning, image segmentation usually adopts unsupervised learning methods such as mean shift [6] and the fractal net evolution algorithm (FNEA) [7]. These algorithms can use only the low-level features of the images, so the resulting segmentation images only have similarities and no semantics.

Hinton *et al.* [8] first proposed AlexNet based on DCNNs, which greatly improves the accuracy of image recognition. DCNNs can be used to extract more abstract features, which is extremely important for semantic segmentation. Farabet *et al.* [9] designed the scene parsing network based on DCNNs by using conditional random fields (CRFs) and patch classification. Compared with the traditional image segmentation method, this method realizes the semantic segmentation of the image. However, because the patch-classification method has many double calculations, which greatly increases the computing overhead of the network and is restricted to the full connection (FC) layer, the network has strict requirements on the size of the input image. Long *et al.* [10] removed

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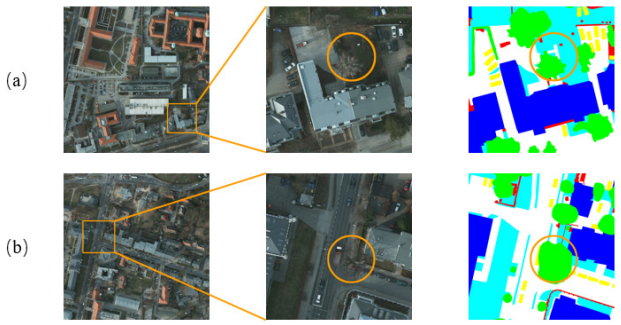


Fig. 1. Example of difficulty in semantic segmentation of remote sensing images. Objects in circles are affected by shadows and occlusion and cannot be properly classified. (a) ISPRS potsdam 4_12. (b) ISPRS potsdam 3_12.

the FC layer in DCNNs and proposed a full convolutional neural network (FCN) based on the encoder–decoder structure, which brings a subversive change to the field of semantic segmentation. It can take an image of any size as an input and process it much faster than the patch-classification method. Since then, almost all semantic segmentation networks based on DCNNs adopt an end-to-end encoder-decoder structure similar to FCN.

Many different modules have been designed to capture the context and rich semantic features of an image. Pyramid scene parsing network (PSPNet) [11] designs the pyramid pooling module (PPM), which combines the characteristics of four different pyramid scales to obtain global context information under different scales. The Atrous Spatial Pyramid Pooling (ASPP) module is proposed in DeeplabV2 [12], which uses atrous convolution to extract global context information by mining features of different scales. Nonlocal network [13] adopts nonlocal operation to capture the long-range dependencies and establish the relationship between any two pixels on the image to obtain the context information.

Compared with natural images, remote sensing images are more easily affected by lighting and camera angles. As shown in Fig. 1(a), road segmentation is difficult owing to tree shadows; (b) vehicle segmentation is difficult because of tree shading. The use of DCNNs for semantic segmentation is not only affected by the data but also by the model itself. On the one hand, the common semantic segmentation method is good for images in the category of big volume, but accurately locating their boundaries is very difficult for the small category. On the other hand, some semantic segmentation method is unable to make full use of context, and so a target is divided into multiple parts.

In addition to the more efficient feature extraction module based on end-to-end structure, many scholars have proposed some improved methods to obtain more accurate semantic segmentation results in high-resolution remote sensing images. Chai *et al.* [14] derived a signed distance map for each semantic class and solved the problem of spatial context learning in semantic segmentation by predicting the distance map of the image instead of the probability of each category. In view of the failure of modeling context spatial relations in remote sensing images, Mou *et al.* [15] constructed spatial relation module and channel relation module to learn and reason the global relationship between any two spatial positions or feature mappings, so as to realize remote spatial relation reasoning. To balance the relationship between feature expression ability and spatial positioning accuracy in semantic segmentation of remote sensing images, Ding *et al.* [16] improved the model by using patch calculation of local attention to enhance the embedding of contextual information and by embedding local focus from high-level features to enrich the semantic information of low-level features.

All the existing state-of-the-art models use high-resolution images as input, but we explore a model that uses a relatively low-resolution image as input to obtain a high-resolution segmentation image.

B. Single Image Super-Resolution

Image super-resolution refers to the recovery of a high-resolution image from a low-resolution image or sequence of images [17]. High resolution means that the pixel density in the image is high and provides more details, and both characteristics are indispensable in many practical applications [18] such as video compression [19] and computerized tomography (CT) image reconstruction [20]. Most of the traditional super-resolution methods adopt interpolation method [21], [22]. They use only the pixel information of the low-resolution image itself, and the pixel points of each position are interpolated based on the information around the pixel, so the reconstructed image is very blurred. Chang *et al.* [23] used local overlap to enhance the compatibility and smoothness constraints between reconstructed high-resolution image blocks. However, the manifold assumption is often inaccurate in practice, so the reconstruction results are usually not ideal. Yang *et al.* [24] proposed a super-resolution method based on sparse representation, which enhances the compatibility between local and global adjacent image blocks. However, an over-complete dictionary is very difficult to train.

With the emergence of deep learning, the single image super-resolution based on DCNNs has become a new solution. In the early DCNNs-based super-resolution method [17], bicubic interpolation [25] was used to restore the low-resolution image to the high-resolution image, and then the feature was extracted by the neural network, and the nonlinear mapping was used to reconstruct the high-resolution image. Kim *et al.* [26] introduced the recursive neural network into the image super-resolution method, which can improve the performance by increasing the recursive depth without adding parameters to the additional convolution. Affected by the gradient, the training of the model is very difficult. At present, the super-resolution network adopts an end-to-end structure [27]–[29]. Sampling operators such as deconvolution [30] and effective subpixel convolution [31] are integrated into a module and placed at the end of the network. The network is convolved on the low-resolution image before up-sampling, which greatly reduces the computation. SRDenseNet [32] introduces DenseNet [33] into a super-resolution problem and uses skip structure to transfer low-level image features to high-level ones, which is used to extract more image features for image reconstruction. LapSRN [34] gradually expands the image resolution while sharing the parameters between

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different pyramid layers and using recursive convolutional network structure, which not only reduces the computational burden but also effectively improves the accuracy. Generative adversarial network for image super-resolution (SRGAN) [35] is a generative adversarial network (GAN) for image super-resolution. Its discriminator controls adversarial loss to make the generated image closer to the real natural image.

Remote sensing images are more complex than natural images. Owing to the large sampling area, usually, a target (e.g., a car or ship) may only occupy a few pixels in the image, which provides less low-level details for the super-resolution network. In addition, the spatial distribution of remote sensing images is more complex, with more low-frequency details and less high-frequency features in the images, and these also result in difficulties in sampling.

To obtain high-quality super-resolution images, existing state-of-the-art models always convolve the resolution of the input original image without using the down-sampling operation, which brings a huge cost to the network. We design a super-resolution module based on the encoder–decoder structure, which can help the model obtain more semantic information and help image super-resolution reconstruction by reducing the amount of computation. This method can not only obtain considerable super-resolution results but also greatly reduce storage requirements and computation.

C. Multitask Learning

Multitask learning (MTL) is a promising field in machine learning, where domain knowledge is used to improve generalization performance through the use of supervisory signals contained in related tasks [36]. In MTL, information among tasks is shared and knowledge is transferred among different tasks. At the same time, owing to the sharing of datasets and underlying characteristics, the generalization performance of the model is enhanced.

Multilabel learning [37] is a form of MTL. MTL also includes multioutput learning, auxiliary training [38], and so on. Lu *et al.* [39] obtained better road detection results by learning road segmentation and centerline extraction. Mask-region-based convolutional neural network (RCNN) [40] accomplishes target detection and instance segmentation by sharing the underlying convolutional layer, which greatly reduces the risk of overfitting.

An important challenge in training multitask models is setting up the loss function. When learning semantic segmentation and super-resolution tasks simultaneously, a naive idea is the loss-weighted sum of different tasks, which is expressed as follows:

$$L_{\text{loss}} = \omega_1 L_{\text{seg}} + \omega_2 L_{\text{sup}}. \tag{1}$$

We employed Deeplabv3+ [41] to test the impact of different weights on the performance of the model on the International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam [42] test set, using F1 score and peak signal to noise ratio (PSNR) to evaluate the performance of semantic segmentation and super-resolution tasks (see Section III-C for details). As shown in Fig. 2 and Table I, network performance is heavily influenced by weights. The cost of manually adjusting these hyperparameters is high, so we had adaptive dynamic weights for different tasks (see Section II-C). When we learned semantic segmentation tasks, super-resolution tasks were treated as auxiliary tasks, and vice versa.

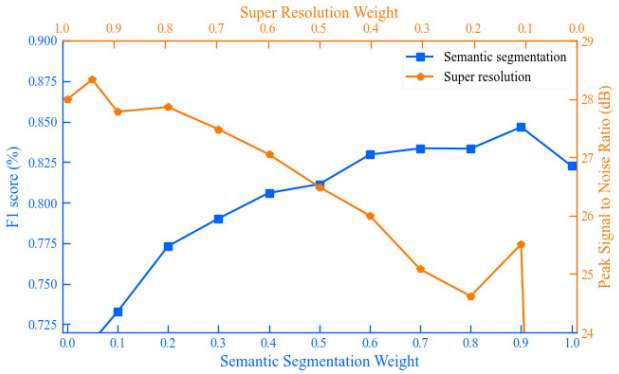


Fig. 2. Illustrations of how MTL improves the performance of individual tasks. By adjusting the weights to make the network more oriented toward learning semantic segmentation tasks or super-resolution tasks, we observed that MTL improves the performance of individual tasks. The results are shown using the ISPRS Potsdam test dataset. The resolution of the input image was 128 × 128 px (0.25×) and was restored to 512 × 512 px. All models used the same experimental setup except for different weights.

TABLE I

COMPARISON OF THE DIFFERENT LOSS WEIGHTS OF SEMANTIC SEGMENTATION AND SUPER RESOLUTION TASKS. DEEPLABV3+ WAS USED TO QUANTIFY THE RESULTS ON THE ISPRS POTSDAM TEST UNDER THE COLNET FRAMEWORK

Task Weights		Metrics	
Semantic Segmentation	Super Resolution	F1 (%)	PSNR (dB)
1	0	82.26	-
0.9	0.1	84.66	25.51
0.8	0.2	83.33	24.62
0.7	0.3	83.35	25.08
0.6	0.4	82.96	26.00
0.5	0.5	81.14	26.49
0.4	0.6	80.60	27.05
0.3	0.7	79.02	27.48
0.2	0.8	77.31	27.86
0.1	0.9	73.27	27.78
0.05	0.95	71.43	28.33
0	1	-	27.99

Inspired by Wang *et al.* [43], we applied super-resolution reconstruction and semantic segmentation to remote sensing images simultaneously from the perspective of MTL. On the basis of sharing encoders and partial decoders, we designed a collaborative network to realize high-resolution image reconstruction and low-resolution image segmentation based on the similarity of image feature structures. We attempted to combine the tasks of semantic segmentation and image super-resolution, which greatly differ, to improve the performance of the network. In summary, the main contributions of this article are as follows.

1) We designed a framework based on the encoder–decoder structure, a ColNet, for simultaneous learning of super-resolution semantic segmentation and super-resolution image reconstruction tasks, which can achieve better performance than the single-task model.

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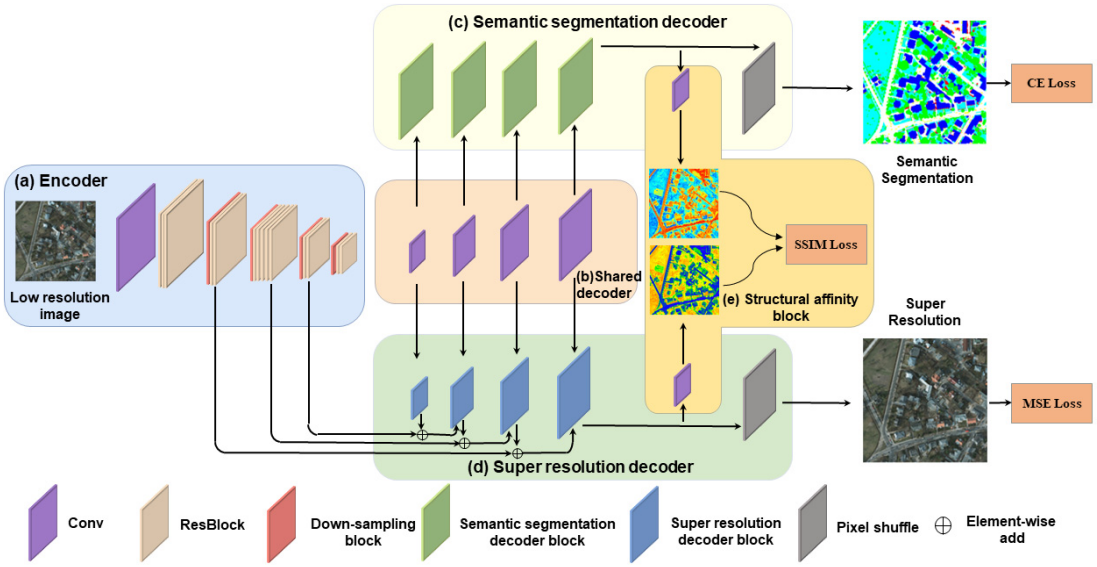


Fig. 3. Overview of the proposed ColNet framework. (a) Encoder, we employ ResNet [48] to extract image features. (b) Shared decoder to realize the hard sharing mechanism of parameters. (c) Semantic segmentation decoder, we use OCR [52] as the semantic segmentation decoder. (d) Super resolution decoder, a progressive decoder using pixel-shuffle [31] reconstructs high-resolution images from features obtained from encoders and shared decoders. (e) SAB relates branches of semantic segmentation and super resolution in SSIM.

2) We demonstrated the versatility of our approach, which can replace arbitrary branches with other state-of-the-art models and achieve significant performance improvements.
3) In MTL, one of the keys is to design the weight loss of each task. To avoid manually adjusting the weight size, we proposed the dynamic weight and verified its effectiveness.
4) Using existing state-of-the-art semantic segmentation and super-resolution networks with the ISPRS Potsdam and Vaihingen datasets [42], [44] as benchmarks, we demonstrated that our proposed framework delivered optimal performance.

The rest of this article is organized as follows. Section II describes in detail the design idea and composition of the proposed ColNet framework. Experimental datasets, model evaluation methods, experimental procedures, and analysis of experimental results are given in Section III. Section IV summarizes the whole article and discusses future work.

II. Methodology

The semantic segmentation model and the super-resolution model are easy to fit the local optimum when learning a single task. Our main motivation is to use MTL to share hidden characteristics to improve the performance of a single model. Based on the above ideas, we proposed a collaborative network for both semantic segmentation and super-resolution tasks. In this section, we first give an overview of the network, and in Section II-B, we further introduce the composition of the network, including semantic segmentation branch (SSB), super-resolution branch (SRB), and structural affinity block (SAB). Finally, in Section II-C, we describe the optimization objectives of the network.

A. Overview of Network Architecture

As shown in Fig. 3, the multitask ColNet framework is composed of two branches, namely, semantic segmentation branches and super-resolution branches. It is a branch structure based on encoder–decoder structure. The network accepts low-resolution images as input and uses SSB and SRB, respectively, for super-resolution segmentation and super-resolution reconstruction. At the same time, SRB is used to realize soft parameter sharing [45] of semantic segmentation decoder and super-resolution decoder.

B. Architecture

1) Encoder: Following the choices of previous work [46], [47], we also selected ResNet [48], pretrained on ImageNet [49], as the backbone. We removed the final pooling layer and the FC layer from ResNet-50, leaving the rest unchanged. In fact, the ColNet framework we proposed can use most of the dominant feature extraction networks today as the backbone, such as DenseNet [33], squeeze-and-excitation network (SENet) [50], or EfficientNet [51].

2) Semantic Segmentation Branch: A typical semantic segmentation network consists of an encoder and a decoder, which encodes the input image and reduces the resolution of the image by down-sampling to reduce the computation, respectively. We chose object-contextual representations network (OCR) [52] as the decoder, which enhances the description of pixel features by learning the relationship between a pixel and object region features according to the rough segmentation results. Instead of an end-to-end approach that uses relational context [13], OCR is a coarse-to-fine approach that learns about the relationship of pixels to regions of an object. First, the target region is learned through the supervision of ground truth. Then, the representation of the target region is calculated by fusing the representation of all pixels on it.

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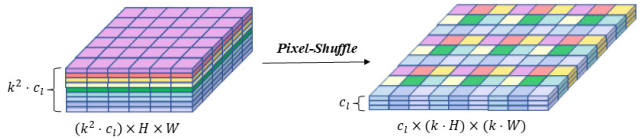


Fig. 4. Pixel shuffle operation with a sampling multiple of k. The dimension of the input feature is $(k^2 \cdot c_l) \times H \times W$, where $k^2 \cdot c_l$ represents the channel dimension, $H \times W$ represents the spatial resolution, and the dimension of the output feature is $c_l \times (k \cdot H) \times (k \cdot W)$.

Finally, the relationship between each pixel and each target region is calculated, and the representation of each region is weighted according to the strength of its relationship to enhance the representation of each pixel.

In our framework, as shown in Fig. 3, seeing that the high-level feature map involves many channels, and the larger space size will significantly increase the calculation cost, we used a shared decoder to further fuse the features of the ResNet output in each stage and then sent the refined features to OCR for decoding the features. The final representation of the feature map that is fed into the OCR module is as below

$$y_i = g(x_i). \tag{2}$$

Here, x_i is the representation of pixel p_i on the feature map obtained by the encoder. $g(\cdot)$ is transformation functions implemented by 1×1 conv, batch normalization (BN), and rectified linear unit (ReLU). Owing to the super-resolution semantic segmentation in the final up-sampling stage, the usual up-sampling factor is very large, such as 8 or 16. Simple sampling methods such as bilinear interpolation do not bring any additional information, so they often cannot achieve good results. Inspired by Shi *et al.* [31], Tong *et al.* [32], and Huang *et al.* [34], we adopted subpixel as the up-sampling method (See Fig. 4). The entire upsampling process is described as

$$I^{\mathrm{SS}} = \mathrm{PS}\left(I^{\mathrm{decoder}}\right) \tag{3}$$

where I^{SS} is the output of the entire SSB, I^{decoder} represents the segmentation result obtained by the OCR decoder and PS is a pixel shuffle operation that rearranges the elements of a $H \times W \times C \cdot r^2$ tensor to a tensor of shape $r \cdot H \times r \cdot W \times C$. Compared with the checkerboard effect [53] caused by deconvolution, subpixel up-sampling expands the feature map by reshaping the spatial dimension and channel dimension, so the checkerboard effect can be avoided.

3) Super-Resolution Branch: As mentioned above, to achieve semantic segmentation and learning at the same time with super-resolution, convolution is carried out at a higher resolution compared with the traditional super-resolution method. The input image is first sampled and encoded, and then the decoder is used for decoding to reduce the computation. As shown in Fig. 5, inspired by u-net (UNet) [54] and the residual network [48], we introduced a decoder for skip connections. High-level features obtain rich detailed information through convolution, expand image resolution through subpixel convolution, and fuse with low-level features after nonlinear parametric ReLU (PReLU) activation function

$$y_l = \varphi(x_l) + \delta(x_{l+1}). \tag{4}$$

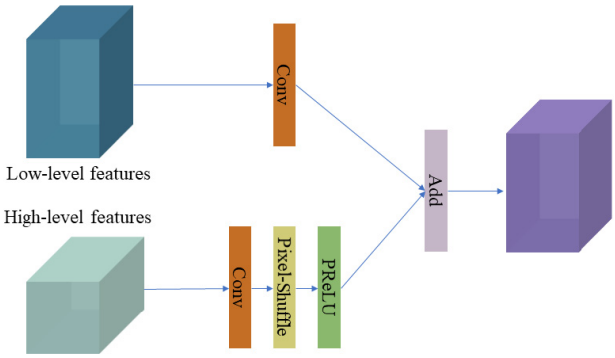


Fig. 5. Details of the super-resolution decoder block component in Fig. 3.

Here, y_l is the feature map obtained by the decoder fusion at lth level, x_l stands for the feature map obtained by the encoder at lth level, and x_{l+1} denotes the higher level of the feature map generated by the encoder. $\varphi(\cdot)$ and $\delta(\cdot)$ are two transformation functions implemented by conv and conv $\rightarrow$ PS$(\cdot)$ $\rightarrow$ PReLU, respectively. Referring to the enhanced deep super-resolution network (EDSR) [55], we removed the BN layer from the decoder because it eliminates network flexibility, which is detrimental to the super-resolution task. Our approach takes advantage of high-level features and low-level details and can significantly reduce computation and memory consumption.

4) Structural Affinity Block: Semantic segmentation usually leads the network to learn more semantic feature information, but the network usually cannot pay much attention to the low-level details. The super-resolution task pays more attention to the detailed features of the image and ignores the high-level semantic features in the feature learning. We designed a feature association module called SAB. It links the features of the semantic segmentation task with those of the super-resolution task through structural similarity (SSIM) [56]. Considering computation and memory overhead, we put SAB in the last stage of feature fusion. At the same time, before the similarity is calculated, the feature is reduced to reduce the amount of calculation. Specifically, for two feature maps whose semantic segmentation input F_{seg} is $H \times W \times C_a$ and super-resolution input F_{sup} is $H \times W \times C_b$, where $H \times W$ means the spatial dimension, C means the channel dimension. As information is also redundant in depth direction, we first used convolution with kernel of 3 to reduce the number of channels in the feature to 1. The SSIM is also calculated for the feature map after dimension reduction. We used the following formula to represent the relationship between the two feature maps:

$$\mathrm{SSIM}(x, y) = \frac{\left(2\mu_x\mu_y + c_1\right)\left(2\sigma_{xy} + c_2\right)}{\left(\mu_x^2 + \mu_y^2 + c_1\right)\left(\sigma_x^2 + \sigma_y^2 + c_2\right)} \tag{5}$$

where μ_x is the average of x; μ_y is the average of y; σ_x is the variance of x; σ_y is the variance of y; σ_{xy} is the covariance of x and y; c is the constant that maintains stability, $c_1 = 0.01^2$, $c_2 = 0.03^2$, respectively; and x and y refer to the convoluted F_{seg} and F_{sup}, respectively.

C. Loss Function

As shown in Fig. 3, the loss of the whole network is composed of three parts, namely, the loss of cross entropy (CE)

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of semantic segmentation, the loss of mean square error (MSE) of the super resolution, and the loss of SSIM of the structure affinity module

$$L_{\text{loss}} = \omega_{\text{ce}} L_{\text{ce}} + \omega_{\text{mse}} L_{\text{mse}} + \omega_{\text{ssim}} L_{\text{ssim}} \tag{6}$$

and

$$L_{\text{ce}}(\theta) = -\sum_{i=1}^{n} x_i \cdot \log(\hat{x}_i) \tag{7}$$

$$L_{\text{mse}}(\theta) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{8}$$

$$L_{\text{ssim}}(\theta) = 1 - \text{SSIM} \tag{9}$$

where x_i represents the semantic segmentation label, $\hat{x}_i$ represents the predicted probability of output from the SSB, y_i is the super-resolution label, and $\hat{y}_i$ is the output value of the SRB.

As shown in Fig. 2, the weight of different task losses in the multitask problem greatly affects the learning direction of the network. In general, the cross-entropy loss of semantic segmentation is more than 100 times that of the loss of super-resolution MSE. Thus, manually adjusting the weight is very difficult. We set different adaptive weights for different tasks and objectives to help network learning. For the semantic segmentation task, we set the weight as

$$\omega_{\text{ce}} = \frac{L_{\text{ce}}}{L_{\text{ce}} + L_{\text{mse}} + L_{\text{ssim}}} \tag{10}$$

$$\omega_{\text{mse}} = \frac{L_{\text{mse}}}{L_{\text{ce}} + L_{\text{mse}} + L_{\text{ssim}}} \tag{11}$$

$$\omega_{\text{ssim}} = \frac{L_{\text{ssim}}}{L_{\text{ce}} + L_{\text{mse}} + L_{\text{ssim}}}. \tag{12}$$

For the super-resolution task, we set the weight as

$$\omega_{\text{ce}} = \frac{L_{\text{mse}} + L_{\text{ssim}}}{L_{\text{ce}} + L_{\text{mse}} + L_{\text{ssim}}} \tag{13}$$

$$\omega_{\text{mse}} = \frac{L_{\text{ce}} + L_{\text{ssim}}}{L_{\text{ce}} + L_{\text{mse}} + L_{\text{ssim}}} \tag{14}$$

$$\omega_{\text{ssim}} = \frac{L_{\text{ce}} + L_{\text{mse}}}{L_{\text{ce}} + L_{\text{mse}} + L_{\text{ssim}}}. \tag{15}$$

III. Experiment

In this section, we describe in detail the datasets used in our experiment, introduce the evaluation indicators, and present the experimental results.

A. Datasets

We benchmarked our proposed ColNets on two challenging state-of-the-art airborne semantic segmentation datasets, namely, ISPRS Potsdam and ISPRS Vaihingen [42]. The ISPRS Potsdam dataset contains 38 patches of 6000 × 6000 in size. Each patch is composed of four channels of red (R), green (G), blue (B), and near-infrared (IR), and each channel has a spectral resolution of 8 bits and a spatial resolution of 5 cm. We chose five patches as the verification set (IDs: 3_13, 4_12, 5_12, 6_12, 7_12), 14 patches as the test set (IDs: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13,5_14, 5_15, 6_13, 6_14, 6_15, 7_13), and the remaining 19 patches as the training set. The corresponding digital surface model (DSM) is not used in the model. The dataset includes six of the most common types of land coverings (Impervious surfaces, Building, Low vegetation, Tree, Car, and Clutter). In general, classification Clutter is not of interest in the semantics of remote sensing images.

Unlike the historic city of Potsdam, which has large buildings, narrow streets, and densely packed settlements, Vaihingen is a relatively small village with many independent small buildings. The ISPRS Vaihingen dataset is composed of 33 patches of various sizes (from 1281 × 2336 px to 2550 × 3816 px). Each patch contains three channels: near-IR, red (R), and green (G), and the sampling distance of the true orthophoto ground is 9 cm. We selected three patches as the verification set (IDs: 7, 13, 21), 13 patches as the training set (IDs: 1, 3, 5, 11,15, 17, 23, 26, 28, 30, 34, 37), and the remaining 17 patches as the test set.

B. Experimental Setup

Our framework is based on Pytorch implementation, with the ResNet-50 [48] pretrained in ImageNet [46] as the backbone. Considering that the images in the Potsdam and Vaihingen datasets were too large to fit into memory, we used a 300 px stride size as a sliding window and clipped the image to 512 × 512 px. We used flipping and multiscale input (scale is [0.5, 0.75, 1.0, 1.25, 1.5, 1.75]) as data augmentation methods to increase the diversity of data. Adam [57] optimizer was adopted for training of all models, with a learning rate of 0.001, $\beta 1 = 0.9$, $\beta 2 = 0.999$, and polynomial learning rate decay strategy was adopted. The learning rate was updated by decay factor $(1 - (\text{iter}/\text{max_iter}))^{0.9}$ after each iteration. To be fair, all the networks were trained with 200 epochs on two GeForce RTX 2080Ti GPUs, and the training started from scratch with open-sourcing code. Besides the backbone of the network, the network was initialized with Kaiming initialization [58]. Other training tricks not mentioned were not used in the model training.

C. Evaluation Metrics

To correctly evaluate the performance of our proposed model, we quantified the performance of the model using four common evaluation metrics of semantic segmentation: overall accuracy (OA), average F1 score (F1), average recall (recall) and, mean Intersection over Union (mIoU), and two image super-resolution evaluation metrics SSIM and PSNR. They are, respectively, calculated as

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$\text{mIoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{18}$$

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{19}$$

$$\text{PSNR}(x, y) = 10\log_{10} \frac{\text{Max}_I^2}{\text{MSE}(x, y)} \tag{20}$$

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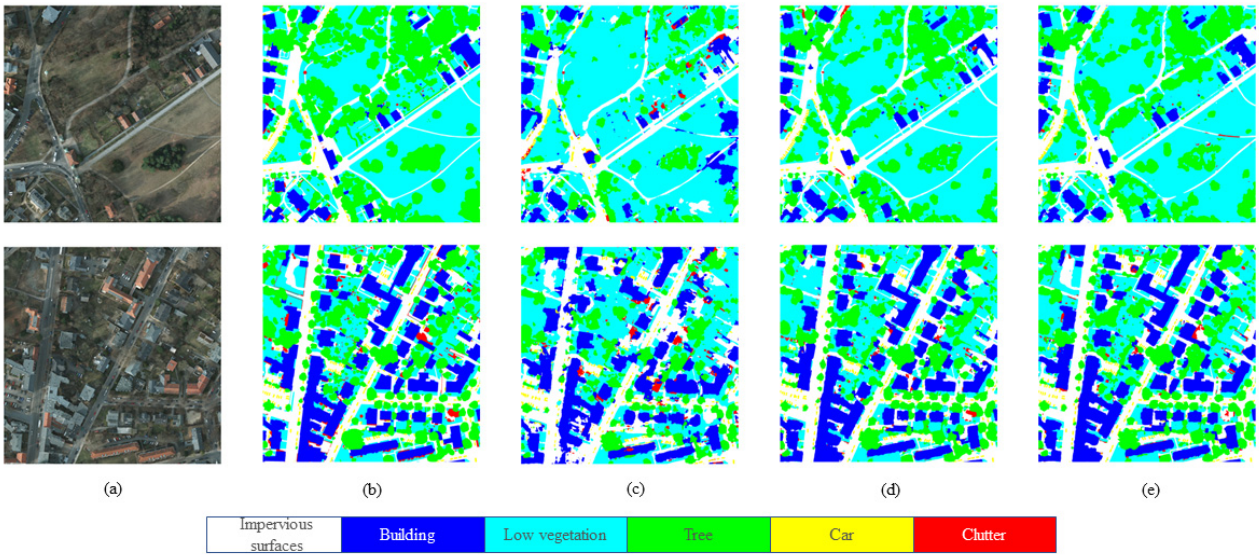


Fig. 6. Visual comparison of semantic segmentation results with input resolution of 128 × 128 on the ISPRS Potsdam test dataset. (a) Input image. (b) Label. (c) Deeplabv3+ processing results. (d) EANet processing results. (e) ColNet processing results.

TABLE II
EXPERIMENTAL RESULTS OF SEMANTIC SEGMENTATION

ISPRS Potsdam Test Data Set
Results without and with ColNet Framework

Input size	**256×256**				**128×128**				**64×64**			
Method	**OA(%) / Recall(%) / mIoU(%) / F1(%)**											
EANet [59]	89.68	**88.60**	80.50	89.01	85.77	84.44	74.27	84.99	75.64	76.83	62.18	75.91
EANet(ours)	**89.86**	88.51	**80.64**	**89.11**	**86.78**	**85.62**	**75.91**	**86.11**	**79.73**	**78.93**	**66.16**	**79.22**
Deeplabv3+ [41]	87.63	86.11	76.91	86.76	83.00	81.78	70.22	82.26	73.10	73.65	58.51	73.10
Deeplabv3+(ours)	**89.53**	**88.05**	**79.97**	**88.70**	**85.61**	**84.04**	**73.77**	**84.66**	**75.91**	**75.51**	**61.43**	**75.58**
BiseNet [60]	**88.62**	**87.63**	**78.92**	**88.09**	85.13	83.30	72.89	84.03	74.95	74.60	61.63	74.70
BiseNet(ours)	87.13	86.35	76.74	86.68	**85.21**	**84.18**	**73.78**	**84.61**	**75.59**	**75.05**	**62.20**	**75.24**
OCR [52]	89.86	**88.53**	80.64	89.14	86.35	85.96	75.87	86.12	83.56	**83.54**	71.93	83.39
OCR(ours)	**90.05**	88.52	**80.80**	**89.22**	**88.49**	**87.19**	**78.50**	**87.77**	**84.38**	83.19	**72.31**	**83.69**

ISPRS Vaihingen Test Data Set
Results without and with ColNet Framework

Input size	**256×256**				**128×128**				**64×64**			
Method	**OA(%) / Recall(%) / mIoU(%) / F1(%)**											
UNet [54]	85.00	86.16	75.40	85.51	74.14	78.62	63.12	75.56	64.93	63.84	54.07	64.37
UNet(ours)	**86.27**	**86.84**	**76.80**	**86.53**	**79.14**	**83.25**	**69.42**	**80.71**	**68.89**	**76.33**	**58.28**	**70.21**
EANet [59]	**81.93**	82.98	71.20	82.41	66.23	**65.24**	55.97	65.71	61.44	60.46	49.40	60.85
EANet(ours)	81.52	**83.74**	**71.30**	**82.48**	**66.31**	65.19	**56.00**	**65.75**	**62.57**	**61.58**	**50.89**	**61.99**
Deeplabv3+ [41]	81.59	**83.32**	70.90	82.35	66.16	74.48	55.19	66.86	60.47	59.69	48.23	59.93
Deeplabv3+(ours)	**81.65**	83.29	**70.96**	**82.38**	**72.46**	**77.28**	**61.48**	**73.98**	**61.01**	**68.83**	**48.96**	**60.52**
OCR [52]	84.46	**86.43**	75.14	85.32	73.81	80.77	64.47	75.56	65.54	**74.84**	54.66	65.71
OCR(ours)	**86.31**	86.11	**76.28**	**86.17**	**79.24**	**81.24**	**68.62**	**80.09**	**68.29**	73.03	**57.02**	**69.35**

where TP, TN, FP, and FN, respectively, refer to true positive, true negative, false positive, and false negative pixels in SSB prediction. MSE(x, y) refers to the mean square deviation loss of prediction and label in the SRB, and Max_I refers to the maximum value of the input image. In this article, we normalized the image, so $\text{Max}_I = 1$.

D. Results

We conducted a series of experiments comparing the ISPRS Potsdam and Vaihingen datasets with state-of-the-art models to verify the validity of our proposed framework. We reported the results of the performance comparison between the ColNet architecture we proposed and the most state-of-the-art semantic segmentation methods at present. We directly reported the ISPRS evaluation guidance metrics F1 Score and OA as well as mIoU and Recall. For image super-resolution reconstruction tasks, we also compared our approach with state-of-the-art models on ISPRS Potsdam and Vaihingen datasets. We directly reported the SSIM and PSNR metrics.

1) Segmentation Proposal Results: Our ColNet design is based on OCR [52]. For the ISPRS Potsdam and Vaihingen test sets, we applied our proposed ColNet framework to several other architectures, including edge-aware network (EANet) [59], Deeplabv3+ [41], BiseNet [60], and UNet [54], and compared the performance.

The results are shown in Table II. We reported three cases when the input image size is 64 × 64 px, 128 × 128 px,

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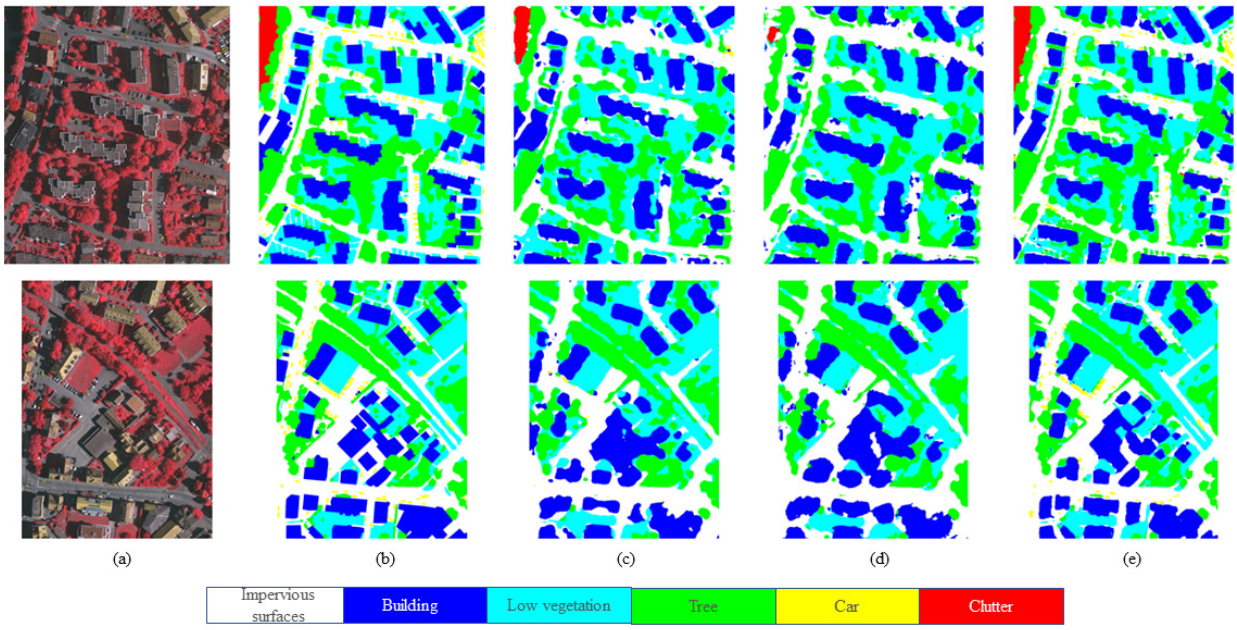


Fig. 7. Visual comparison of semantic segmentation results with input resolution of 128 × 128 on the ISPRS Vaihingen test dataset. (a) Input image. (b) Label. (c) Deeplabv3+ processing results. (d) EANet processing results. (e) ColNet processing results.

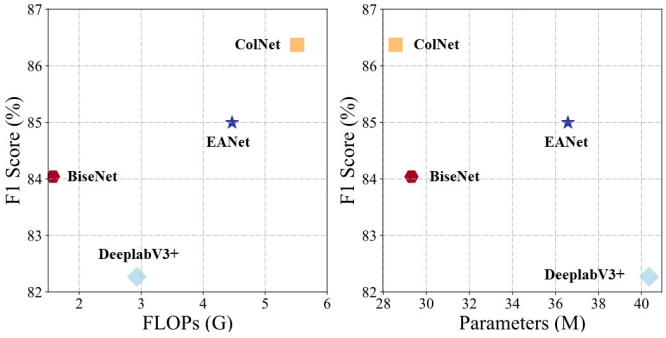


Fig. 8. Comparison of actual total costs between different models during testing. RTX 2080 Ti was evaluated on a single node. The number of arguments and FLOPs were tested using an open-source package for a single image with a resolution of 128 × 128.

TABLE III
COMPARISON OF COMPUTATIONAL COMPLEXITY AND NUMBER OF ARGUMENTS. WE TESTED THE INFERENCE TIME 100 TIMES AND AVERAGED IT

Method	Parameters (M)	Inference Time (ms)	FLOPs (G)
BiseNet	29.30	5.681	1.578
DeepLabv3+	40.35	4.637	2.932
EANet	36.58	16.49	4.466
ColNet	28.57	4.722	5.517

TABLE IV
EXPERIMENTAL RESULTS OF SUPER-RESOLUTION

Symbol	Data Set	×2	×4	×8
		SSIM/PSNR	SSIM/PSNR	SSIM/PSNR
Bicubic	Potsdam Val Data Set	0.8836/31.76	0.7824/28.39	0.7161/24.75
EDSR[55]		0.9036/32.69	0.7866/28.13	0.7166/25.35
MSRN[61]		0.8929/32.33	0.7627/28.24	0.7121/24.77
DDBPN[62]		0.8946/32.18	0.7834/27.15	0.7155/24.92
RCAN[63]		0.9043/32.88	0.7798/28.10	0.7145/25.42
Ours		**0.9289/34.05**	**0.7978/28.66**	**0.7222/25.78**
Bicubic	Potsdam Test Data Set	0.8830/32.38	0.7758/28.45	0.7016/25.54
EDSR[55]		0.8971/32.90	0.7758/28.55	0.7058/25.84
MSRN[61]		0.8844/32.12	0.7743/27.09	0.7008/25.36
DDBPN[62]		0.8864/32.40	0.7721/27.78	0.7046/25.54
RCAN[63]		0.9052/33.14	0.7745/28.44	0.7046/25.91
Ours		**0.9278/34.64**	**0.7898/29.21**	**0.7117/26.32**
Bicubic	Vaihingen Val Data Set	0.8718/29.77	0.7245/24.81	0.6102/21.97
EDSR[55]		0.8681/29.91	0.7223/25.63	0.5950/22.17
DDBPN[62]		0.8709/29.92	0.7391/25.68	0.6202/22.53
MSRN[61]		0.8748/30.22	0.7282/25.48	0.6199/22.44
SRDenseNet [32]		0.8795/29.63	0.7351/25.42	0.5819/19.03
Ours		**0.8821/30.32**	**0.7436/26.05**	**0.6255/22.63**
Bicubic	Vaihingen Test Data Set	0.8647/28.75	0.7195/24.74	0.6010/21.02
EDSR[55]		0.8738/29.92	0.7342/25.65	0.6139/22.28
DDBPN[62]		0.8772/29.97	0.7512/25.76	0.6397/22.65
MSRN[61]		0.8811/30.30	0.7408/25.59	0.6393/22.56
SRDenseNet [32]		0.8856/29.75	0.7484/25.53	0.6060/19.46
Ours		**0.8878/30.35**	**0.7553/26.08**	**0.6448/22.75**

and 256 × 256 px, and their output image resolution is 512 × 512 px. For low-resolution images of different sizes, our proposed OCR-based ColNet always yields the best performance. EANet is a very strong competitor on the ISPRS Potsdam dataset, and Figs. 6 and 7 show the qualitative evaluation results of semantic segmentation. The results show that our method is satisfactory.

Fig. 8 and Table III show the relationship between the performance of different models and resource consumption. In the Potsdam Test dataset, the resolution of the input image is

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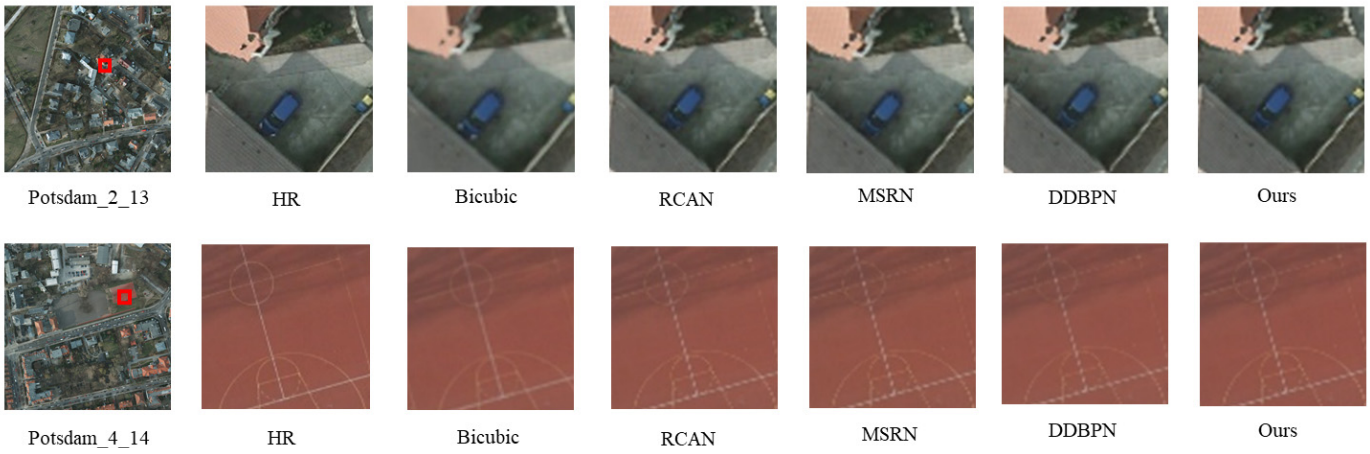


Fig. 9. Super-resolution reconstructed images generated by different state-of-the-art models on the ISPRS Potsdam dataset with a sampling factor of 4.

TABLE V
TO EVALUATE THE EFFECT OF IMAGE INPUT RESOLUTION ON THE NETWORK ON THE POTSDAM VERIFICATION SET, WE REPORT THE MIOU (%)

Method	Input size	Output size	mIoU (%)
DeepLabv3+ [41]	512×512	512×512	84.5
	256×256	256×256	77.3
	128×128	128×128	71.8
DeepLabv3+(our)	256×256	512×512	81.1
	128×128	512×512	76.2
EANet [59]	512×512	512×512	83.8
	256×256	256×256	77.8
	128×128	128×128	72.2
EANet (our)	256×256	512×512	82.0
	128×128	512×512	77.1

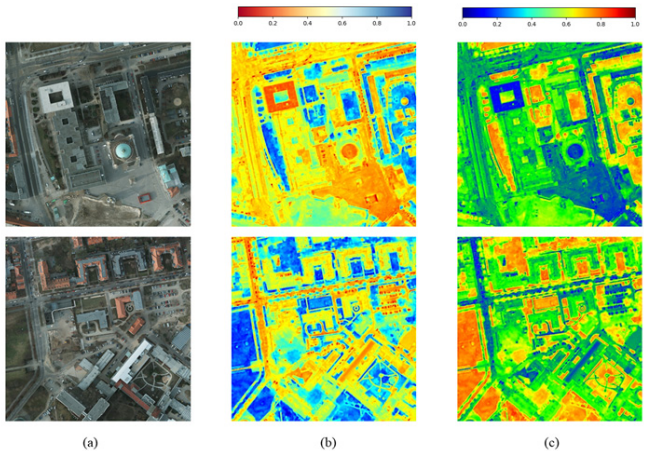


Fig. 10. SSB and SRB visually display features with input resolution of 128 × 128 px. (a) Origin image. (b) SSB feature visualization. (c) SRB feature visualization.

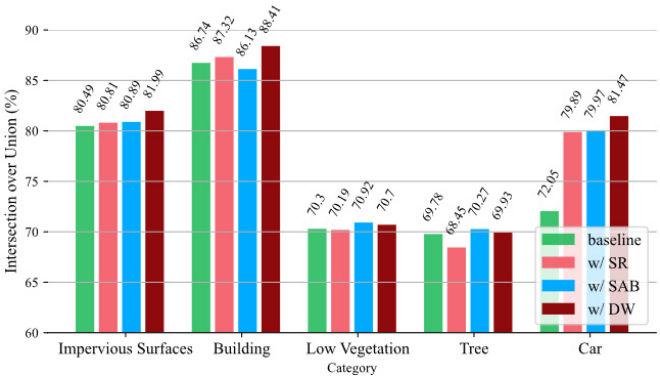


Fig. 11. Effectiveness of each type of IoU scoring on the Potsdam test set in the ColNet framework.

128 × 128 px. Our proposed network has the best performance on the F1 Score, with the highest floating point operations per seconds (FLOPs), the least number of arguments and the medium reasoning speed, which are 5.517 G, 28.57 M, and 4.722 ms, respectively.

The same model performs slightly less well than Potsdam on the Vaihingen dataset. We think of two possible reasons. First, the number of remote sensing images in the Vaihingen dataset is much smaller than that of Potsdam. Second, each image in Potsdam is made up of four channels that provide more information for the network, compared with three in Vaihingen, making it harder for the network to learn.

2) Super-Resolution Proposal Results: Quantitative and qualitative experiments on Potsdam and Vaihingen datasets show that our approach is effective for image super-resolution reconstruction tasks. We used advanced super-resolution reconstruction models, including EDSR [55], multi-scale residual network (MSRN) [61], dense deep back-projection network (DDBPN) [62], very deep residual channel attention network (RCAN) [63], and SRDenseNet [32], to compare with our methods.

We conducted experiments with sampling factors of ×2, ×4, and ×8. Table IV reported SSIM and PSNR scores in each case. The qualitative results in Fig. 9 show that our model achieves satisfactory results. Our method achieves the highest score in all cases. At the same time, the image boundary reconstructed by our method is relatively clear. On the contrary, the distortion and blur of other super-resolution models are more serious.

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TABLE VI

LOCAL EVALUATIONS OF EACH CATEGORY ON THE ISPRS POTSDAM TEST SET. WE REPORT THE IOU AND F1 SCORES

Method	Impervious Surfaces		Building		Low Vegetation		Tree		Car		Average	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
Baseline	80.49	89.19	86.74	92.90	70.30	82.56	69.78	82.20	72.05	83.76	75.87	86.12
+SRB	80.81	89.39	87.32	93.23	70.19	82.48	68.45	81.27	79.89	88.82	77.33	87.03
+SAB	80.89	89.44	86.13	92.55	**70.92**	**82.98**	**70.27**	**82.54**	79.98	88.88	77.63	87.27
+Dynamic Weight	**81.99**	**90.10**	**88.41**	**93.85**	70.70	82.83	69.93	82.30	**81.47**	**89.79**	**78.50**	**87.77**

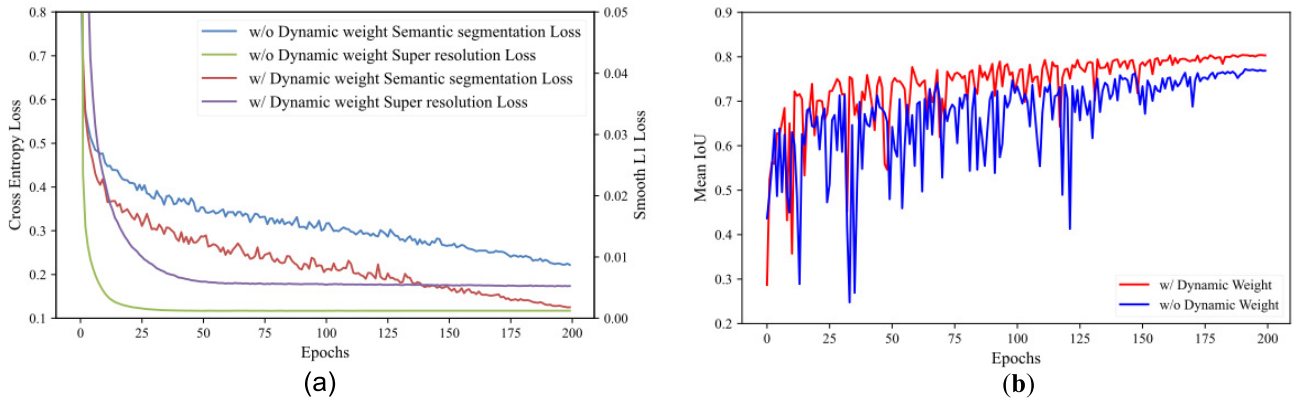


Fig. 12. Compare the effects of using dynamic weights on network performance. When dynamic weights are not used, the weights of CE loss and smooth L1 loss are 1. (a) Convergence relationship between epochs and loss values. (b) Relationship between epochs and mIoU. Performance using dynamic weights was significantly superior.

E. Ablation Study

We first study the influence of input image resolution on the proposed algorithm. We use bicubic interpolation to subsample the original image and interpolate the nearest neighbor of the label. The experiment is performed on Deeplabv3+ [41] and EANet [59], and the mIoU is evaluated on Potsdam Validation set. As shown in Table V, taking EANet as an example, the resolution of the input image is halved and Intersection over Union (IoU) performance is reduced by only 1.8%.

To further test the efficiency of our model, we designed a series of ablation experiments. The ablative study is to test the importance and performance of each part of the model. For the four parts of our model, SSB, SRB, feature correlation module, and loss dynamic weight, we conducted ablation research.

For the semantic segmentation task, we still used ResNet-50 + OCR as the baseline to experiment on the ISPRS Potsdam test set. The input spatial resolution of the baseline is 128 × 128, and the spatial resolution after super-resolution semantic segmentation is 512 × 512. We first added the SRB, then added the SAB, and added the linear sum of the losses per task. Fig. 10 shows the specific diagnostic visualization results of SSB and super-resolution reconstruction branch after using SAB. Finally, we verified the validity of dynamic weights. As shown in Table VI, we reported the IoU and F1 scores for each category in the ablation experiment of the semantic segmentation task. The results show that our method performs well.

Fig. 11 illustrates the effectiveness of our proposed approach. For regular object classes such as Impervious Surfaces, buildings, and cars, our ColNet framework shows significant improvements over baseline in IoU scores, by 1.5%, 1.67%, and 9.42%, respectively.

TABLE VII

IN THE POTSDAM TEST SET TO EVALUATE THE SUPER-RESOLUTION RECONSTRUCTION PERFORMANCE, THE EFFECTS OF DIFFERENT COMPONENTS OF OUR METHOD ON THE RESULTS WERE OBTAINED BY FACTOR 4

Method	Metrics	
	SSIM	PSNR
Baseline	0.7633	27.99
+SSB	0.7694	28.33
+SAB	0.7703	28.38
+Dynamic Weight	0.7898	29.21

For the image super-resolution reconstruction task, we also carried out a similar ablation experiment. Taking ColNet's SRB as the baseline, we conducted a sampling factor of 4 tests on the ISPRS Potsdam Test Set. We added semantic segmentation branches, structural association modules, and dynamic weights. Table VII reports the test results, and our approach significantly improves the baseline performance by 0.0265 and 1.22 dB on SSIM and PSNR, respectively.

To demonstrate the importance of our proposed dynamic weights, we conducted experiments using OCR [52] in the ISPRS Potsdam validation set. The resolution of the input image is 128 × 128, and super-resolution reconstruction image and super-resolution semantic segmentation image with 512 × 512 resolution are obtained through SSB and SRB, respectively. Since the image super-resolution reconstruction

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task is simpler than the semantic segmentation task, the SRB of the network converges at the early stage of training, and the higher loss weight will affect the learning of the encoders and decoders shared by the network. As shown in Fig. 12(a), the cross-entropy loss is not in the same order of magnitude as the Smooth L1 loss. Without the use of dynamic weights to optimize the target with the same weights, it can be seen that the SSB converges at about 15 epochs, which may lead to overfitting of shallow shared modules. After the use of dynamic weights, the convergence speed of super-resolution branches is delayed and starts to converge at about 40 epochs, which can prevent network overfitting. The results in Fig. 12(b) also verify this problem. The use of dynamic weights can reduce the excessive weight and make the influence of convergence of super-resolution branches on semantic segmentation.

IV. Conclusion

In this article, we propose our ColNet framework for super-resolution semantic segmentation and image super-resolution reconstruction. Specifically, our framework is based on an encoder–decoder structure, consisting of a shared encoder, three decoders, and SAB. Compared with a single model trained for each task, our framework can learn different features of different task focuses to improve the generalization performance of the model and achieve state-of-the-art performance. Our framework is also generic and can replace any branch with other state-of-the-art models to improve the performance of the single-task model. In addition, we design the dynamic weight of multitask loss. Ablation experiments on the ISPRS Potsdam test set are conducted to demonstrate the effectiveness of the method. For sampling factor 4 and for semantic segmentation tasks, our model improves by 2.63% and 1.65% compared with baseline in IoU and F1 metrics. For image super-resolution reconstruction tasks, our model improves the SSIM and PSNR metrics by 0.0265 and 1.22 over the baseline.

As shown in Fig. 2, for MTL, the weight of different task losses greatly affects the performance of the model, and the dynamic weight we designed may not be the optimal solution. For future work, we will continue to look at how to quantify the weights between different tasks.

References

[1] R. O. Green *et al.*, "Imaging spectroscopy and the airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 227–248, Sep. 1998.

[2] C. S. Murthy, S. Thiruvengadachari, P. V. Raju, and S. Jonna, "Improved ground sampling and crop yield estimation using satellite data," *Int. J. Remote Sens.*, vol. 17, no. 5, pp. 945–956, Mar. 1996.

[3] L. L. Grewe and R. R. Brooks, "Atmospheric attenuation reduction through multisensor fusion," *Proc. SPIE*, vol. 3376, pp. 102–109, Mar. 1998.

[4] X. Shi, X. Li, and R. Zhu, "A new approach of generating atmospheric path radiation images based on atmospheric radiation transmittance theory," in *Proc. Int. Conf. Comput. Appl. Syst. Modeling (ICCASM)*, vol. 4, Oct. 2010, pp. V4-559–V4-562.

[5] M. Lan, Y. Zhang, L. Zhang, and B. Du, "Global context based automatic road segmentation via dilated convolutional neural network," *Inf. Sci.*, vol. 535, pp. 156–171, Oct. 2020.

[6] D. Comaniciu and P. Meer, "Mean shift: A robust approach toward feature space analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 5, pp. 603–619, May 2002.

[7] U. C. Benz, P. Hofmann, G. Willhauck, I. Lingenfelder, and M. Heynen, "Multi-resolution, object-oriented fuzzy analysis of remote sensing data for GIS-ready information," *ISPRS J. Photogramm. Remote Sens.*, vol. 58, nos. 3–4, pp. 239–258, Jan. 2004.

[8] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. 25th Int. Conf. Neural Inf. Process. Syst. (NIPS)*. Red Hook, NY, USA: Curran Associates, 2012, pp. 1097–1105.

[9] C. Farabet, C. Couprie, L. Najman, and Y. LeCun, "Learning hierarchical features for scene labeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1915–1929, Aug. 2013.

[10] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[11] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 6230–6239.

[12] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.

[13] X. Wang, R. B. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Salt Lake City, UT, USA, Jun. 2018, pp. 7794–7803.

[14] D. Chai, S. Newsam, and J. Huang, "Aerial image semantic segmentation using DCNN predicted distance maps," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 309–322, Mar. 2020.

[15] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[16] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[17] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2015.

[18] S. Zhang, Q. Yuan, J. Li, J. Sun, and X. Zhang, "Scene-adaptive remote sensing image super-resolution using a multiscale attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4764–4779, Jul. 2020.

[19] L.-W. Kang, B.-C. Chuang, C.-C. Hsu, C.-W. Lin, and C.-H. Yeh, "Self-learning-based single image super-resolution of a highly compressed image," in *Proc. IEEE 15th Int. Workshop Multimedia Signal Process. (MMSP)*, Sep. 2013, pp. 224–229.

[20] Y. Zhang, Y. Xi, Q. Yang, W. Cong, J. Zhou, and G. Wang, "Spectral CT reconstruction with image sparsity and spectral mean," *IEEE Trans. Comput. Imag.*, vol. 2, no. 4, pp. 510–523, Dec. 2016.

[21] L. Zhang and X. Wu, "An edge-guided image interpolation algorithm via directional filtering and data fusion," *IEEE Trans. Image Process.*, vol. 15, no. 8, pp. 2226–2238, Aug. 2006.

[22] K. W. Hung and W. C. Siu, "Robust soft-decision interpolation using weighted least squares," *IEEE Trans. Image Process*, vol. 21, no. 3, pp. 1061–1069, Mar. 2012.

[23] H. Chang, D.-Y. Yeung, and Y. Xiong, "Super-resolution through neighbor embedding," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Washington, DC, USA, Jul. 2004, pp. 275–282.

[24] J. Yang, J. Wright, T. Huang, and Y. Ma, "Image super-resolution as sparse representation of raw image patches," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2008, pp. 24–26.

[25] R. G. Keys, "Cubic convolution interpolation for digital image processing," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 29, no. 6, pp. 1153–1160, Dec. 1981.

[26] J. Kim, J. K. Lee, and K. M. Lee, "Deeply-recursive convolutional network for image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 1637–1645.

[27] C. Dong, C. C. Loy, and X. Tang, "Accelerating the super-resolution convolutional neural network," in *Proc. 14th Eur. Conf. Comput. Vis.*, Amsterdam, The Netherlands, Oct. 2016, pp. 391–407.

[28] W. Yang *et al.*, "Deep edge guided recurrent residual learning for image super-resolution," *IEEE Trans. Image Process.*, vol. 26, no. 12, pp. 5895–5907, Dec. 2017.

[29] S. Mei, R. Jiang, X. Li, and Q. Du, "Spatial and spectral joint super-resolution using convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4590–4603, Jul. 2020.

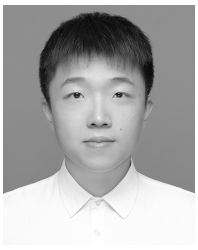
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[30] M. D. Zeiler, G. W. Taylor, and R. Fergus, "Adaptive deconvolutional networks for mid and high level feature learning," in *Proc. Int. Conf. Comput. Vis.*, Barcelona, Spain, Nov. 2011, pp. 2018–2025.
[31] W. Shi *et al.*, "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 1874–1883.
[32] T. Tong, G. Li, X. Liu, and Q. Gao, "Image super-resolution using dense skip connections," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 4809–4817.
[33] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 4700–4708.
[34] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Fast and accurate image super-resolution with deep Laplacian pyramid networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 11, pp. 2599–2613, Nov. 2019.
[35] C. Ledig *et al.*, "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 4681–4690.
[36] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 7482–7491.
[37] M.-L. Zhang and Z.-H. Zhou, "A review on multi-label learning algorithms," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 8, pp. 1819–1837, Aug. 2014.
[38] Y. Zhang and Q. Yang, "A survey on multi-task learning," 2017, *arXiv:1707.08114*. [Online]. Available: http://arxiv.org/abs/1707.08114
[39] X. Lu *et al.*, "Multi-scale and multi-task deep learning framework for automatic road extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9362–9377, Nov. 2019.
[40] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 2961–2969.
[41] L. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. 15th Eur. Conf. Comput. Vis.*, Munich, Germany, Sep. 2018, pp. 801–818.
[42] *ISPRS 2D Semantic Labeling Contest*. Accessed: Jul. 2, 2018. [Online]. Available: http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html
[43] L. Wang, D. Li, Y. Zhu, L. Tian, and Y. Shan, "Dual super-resolution learning for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 3773–3782.
[44] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the Art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.
[45] T. Evgeniou, C. A. Micchelli, and M. Pontil, "Learning multiple tasks with kernel methods," *J. Mach. Learn. Res.*, vol. 6, pp. 615–637, Apr. 2005.
[46] T. Pohlen, A. Hermans, M. Mathias, and B. Leibe, "Full-resolution residual networks for semantic segmentation in street scenes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 4151–4160.
[47] H. Zhao *et al.*, "PSANet: Point-wise spatial attention network for scene parsing," in *Proc. 15th Eur. Conf. Comput. Vis.*, Munich, Germany, Sep. 2018, pp. 267–283.
[48] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.
[49] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Miami, FL, USA, Jun. 2009, pp. 20–25.
[50] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 7132–7141.
[51] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, Long Beach, CA, USA, Jun. 2019, pp. 6105–6114.
[52] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. 16th Eur. Conf.*, Glasgow, U.K., Aug. 2020, pp. 173–190.
[53] A. Odena, V. Dumoulin, and C. Olah. *Deconvolution and Checkerboard Artifacts*. Accessed: Oct. 17, 2016. [Online]. Available: http://distill.pub/2016/deconv-checkerboard
[54] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," Medical Image Computing and Computer-Assisted Intervention - MICCAI 2015.
[55] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Honolulu, HI, USA, Jul. 2017, pp. 136–144.
[56] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[57] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, May 2015. [Online]. Available: https://dblp.org/rec/journals/corr/KingmaB14.html?view=bibtex
[58] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 1026–1034.
[59] G. Yang, Q. Zhang, and G. Zhang, "EANet: Edge-aware network for the extraction of buildings from aerial images," *Remote Sens.*, vol. 12, no. 13, p. 2161, Jul. 2020.
[60] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiSeNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. 15th Eur. Conf.*, Munich, Germany, Sep. 2018, pp. 325–341.
[61] J. Li, F. Fang, K. Mei, and G. Zhang, "Multi-scale residual network for image super-resolution," in *Proc. 15th Eur. Conf.*, Munich, Germany, Sep. 2018, pp. 517–532.
[62] M. Haris, G. Shakhnarovich, and N. Ukita, "Deep back-projection networks for super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Salt Lake City, UT, USA, Jun. 2018, pp. 1664–1673.
[63] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proc. 15th Eur. Conf.*, Munich, Germany, Sep. 2018, pp. 286–301.



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